

Optimal Capital Structure (Session 8)

By the end of this session you must know why enterprise value is maximized when the WACC is minimized, how rising leverage moves the cost of debt, the cost of equity and the tax shield, what the U-shaped WACC curve and the optimal leverage ratio D/E^* mean, why real companies do not sit at the theoretical optimum, how debt appetite changes across the corporate life cycle, the four debt strategies firms adopt, and how to build the operating cash flow (FCFO) that determines a firm's financing need.

Plan

1. The WACC logic: maximize value by minimizing WACC
2. How leverage moves k_d , k_e and the WACC
3. The U-shaped curve and the optimal leverage D/E^*
4. Why real firms do not sit at the optimum
5. Debt across the corporate life cycle
6. The four debt strategies
7. Managing capital structure: the objective and the FCFO build
8. Common Mistakes & Exam Traps
9. Must Know Absolutely

1. The WACC logic: maximize value by minimizing WACC

- Under the DCF approach, a firm's enterprise value (EV) is the present value of its operating cash flows (FCFO) discounted at the weighted average cost of capital (WACC).
- Holding operating cash flows constant, **EV is maximized exactly when the WACC is minimized** — the whole capital-structure question is therefore "what mix of debt and equity gives the lowest WACC?".
- Management acts on the capital-structure **weights** (the mix of debt and equity); the cost of each component, k_d and k_e , adjusts as a consequence.

$$WACC = [D / (D+E)] \times k_d \times (1 - t) + [E / (D+E)] \times k_e$$

- where:
- **k_d** — cost of debt
- **k_e** — cost of equity (from the CAPM)
- **t** — marginal tax rate (the tax shield)
- **D, E** — market values of debt and equity

What this actually does, step by step:

- Compute the weight of debt $D/(D+E)$ and of equity $E/(D+E)$ in total financing.
- Charge debt at its **after-tax** cost $k_d(1-t)$, because interest is tax-deductible.
- Charge equity at k_e (no tax adjustment — dividends are not deductible).
- Add the two weighted costs: the result is the single rate used to discount all of the firm's operating cash flows.

$$EV = \sum_{t=1.. \infty} FCFO_t / (1 + WACC)^t$$

- where **$FCFO_t$** is the operating cash flow of year t . A lower WACC raises every discounted term, so EV rises.

2. How leverage moves kd, ke and the WACC

As leverage rises, three forces pull on the WACC — two push it up, one pulls it down.

- **Cost of debt: kd rises.** Higher leverage raises the probability of default; lenders demand a wider credit spread.
- **Cost of equity: β and ke rise.** Financial risk shifts onto shareholders, so the levered beta rises; through the CAPM a higher βL lifts ke.

$$\beta_L = \beta_U \times [1 + (1 - t) \times D/E]$$

- where:
- **βL** — levered beta (equity beta of the indebted firm)
- **βU** — unlevered beta (the pure business risk, before financing)
- **t** — tax rate; **D/E** — the leverage ratio

What this actually does: it starts from the firm's operating risk (βU), then scales it up in proportion to how much debt sits on top of each euro of equity. More D/E means each shareholder carries more of the firm's fixed obligations, so the equity becomes riskier and ke increases.

- **Tax shield and cheaper weight: WACC falls.** Interest is tax-deductible, so debt is charged at $k_d(1-t)$; on top of that, leverage increases the share of the cheaper source of financing (typically $k_d < k_e$).

Net effect: because the shield offsets the rising costs at first, the WACC is **non-linear** — it first falls, reaches a minimum, then rises once leverage passes the optimal point.

3. The U-shaped curve and the optimal leverage D/E*

Reading the WACC curve (WACC % on the vertical axis, leverage D/E on the horizontal axis):

- **Left of D/E*:** the marginal benefit of the tax shield exceeds the marginal cost of financial distress — the WACC falls.
- **At D/E*:** the WACC is minimized and, equivalently, enterprise value is maximized.
- **Right of D/E*:** distress and agency costs dominate the shield — the WACC rises again.

Core message: enterprise value is the mirror image of the WACC — EV peaks at precisely the leverage where the WACC bottoms out.

Watch out: the numeric "ABC example" behind the curve is shown as a table/chart in the slides; its figures are not recoverable from the text extraction. The logic (U-shape, optimum at D/E*) is what the exam expects.

4. Why real firms do not sit at the optimum

The core logic holds — a lower WACC means a higher valuation, **but only if it is sustainable**. Markets punish a low WACC bought with excessive risk.

Real-world frictions that matter more than the "optimal point":

- **Market timing** — firms issue equity when valuations are high and debt when rates are low.

- **Financial flexibility** — keeping unused debt capacity is often worth more than minimizing today's WACC.
- **Ratings and covenants** — slipping below investment grade can sharply raise the cost of funding.
- **Management incentives** — executives may prefer lower leverage to limit distress (and career) risk.

What firms actually do:

- Target a **credit rating** (e.g. BBB / investment grade).
- Operate within a **leverage range**, not a single point.
- Adjust **opportunistically** as market conditions change.

5. Debt across the corporate life cycle

Financing needs — and the appetite for leverage — shift as a company matures. Increasing cash-flow stability means increasing debt capacity.

Phase	Cash-flow profile	Typical funding
1. Start-up	Negative cash flows, high uncertainty	Equity: business angels, venture capital
2. Development	Scaling revenue, still cash-hungry	Venture and development capital
3. Maturity	Stable, predictable cash flows — debt capacity opens up	Bank financing, private equity / LBOs
4. Decline	Falling cash flows	Turnaround and restructuring funds

Core message: the leveraged buyout lives in the **maturity** phase — where stable cash flows can safely carry debt.

6. The four debt strategies

Firms position themselves along two axes: the level of leverage and its strategic purpose.

- **Low leverage** (little or no debt; e.g. Armani, Apple, Microsoft). Why: volatile cash flows would jeopardize debt service; liquidity preserves flexibility through turbulence; zero debt signals financial strength to investors.
- **Ordinary debt** (working capital / debt > 50%). Mostly short-term funding. Why: a deliberate short-term focus (flexibility, no long-term commitments), or limited access to long-term debt (weak credit profile, early-stage model, high long rates, thin collateral).
- **Debt for growth** (capex / debt > 50%; NFP / net fixed assets > 50%). Leverage funds long-term programs such as M&A, where future cash generation is predictable enough to support it.
- **Aggressive leverage** (NFP / EBITDA > 3.0×; D/E above the industry average). High debt to win share (aggressive pricing, capacity, marketing, acquisitions) or to maximize a sponsor's IRR — **the LBO zone**.

7. Managing capital structure: the objective and the FCFO build

The objective, in two parts:

- **Meet financial needs** — fund the maintenance and development of the business, starting from the operating cash flow it generates (FCFO).
- **Avoid financial tension** — keep incoming cash flows aligned with debt maturities, monitored through sustainability indicators.

$$\text{FCFO} = \text{EBIT} - \text{Taxes on EBIT} + \text{D\&A} - \Delta\text{Working capital} - \text{Net capex}$$

- where:
- **EBIT – Taxes on EBIT = NOPAT** (both from the income statement)
- **D&A** — depreciation and amortization (added back, non-cash; from the IS)
- **$\Delta\text{Working capital}$** — from the balance sheet; a use of cash if it grows, a source if it shrinks
- **Net capex** — investment net of divestitures, from the balance sheet

What this actually does, step by step:

- Start from operating profit (EBIT) and take out the taxes on it: that gives NOPAT.
- Add back D&A because it reduced profit without consuming cash.
- Subtract the cash absorbed by growing working capital.
- Subtract net investment in fixed assets.
- The result is the cash the operations themselves generate — the money available to service debt and fund growth.

Where the financial need comes from — three questions:

- **Self-financing?** Is FCFO sufficient to cover the plan, or not?
- **Working capital** — does it absorb cash (a use) or release it (a source)?
- **Investment** — how much do capex and divestitures absorb or generate?

The same FCFO build, read across firm types:

- **Self-sufficient firm** — high NOPAT, low ΔWC , low capex: FCFO comfortably covers needs; a cash generator with at most temporary needs.
- **Growing firm** — positive but medium NOPAT, high ΔWC , high capex: expansion creates a structural financing need.

Core message: profitability alone does not determine the financing need — working-capital dynamics and the investment programme decide whether a firm funds itself or must raise external capital.

Watch out: the worked "Hippocrates Holding" example is a numeric table in the slides; its figures are not recoverable from the text extraction.

8. Common Mistakes & Exam Traps

- Confusing the direction of causality: management chooses the **weights**, and k_d and k_e **adjust as a consequence** — not the other way around.

- Forgetting the tax adjustment on debt in the WACC: debt enters at $k_d(1-t)$, equity at k_e with no adjustment.
- Believing more debt always lowers the WACC: past D/E^* , distress and agency costs dominate the shield and the WACC rises again.
- Treating the optimal D/E **as an operational target: real firms manage a leverage range and a target rating***, not a single point.
- Placing LBOs in the wrong life-cycle phase: they belong to **maturity**, where cash flows are stable — not to start-ups or decline.
- Reading FCFO as profit: NOPAT is not cash until D&A, working capital and capex are taken into account.

9. Must Know Absolutely

Must know:

- EV is maximized when the WACC is minimized; management moves the D/E weights and k_d , k_e adjust.
- $WACC = D/(D+E) \cdot k_d \cdot (1-t) + E/(D+E) \cdot k_e$.
- Leverage pushes k_d up (default risk), k_e up ($\beta_L = \beta_U[1+(1-t)D/E]$), and pulls the WACC down via the tax shield — hence the U-shape and the optimum D/E^* .
- Real firms target a rating and a leverage range, not the theoretical optimum (timing, flexibility, ratings, incentives).
- Debt capacity grows with cash-flow stability across the life cycle; LBOs live in the maturity phase.
- Four debt strategies: low leverage, ordinary debt, debt for growth, aggressive leverage ($NFP/EBITDA > 3.0\times = \text{LBO zone}$).
- $FCFO = EBIT - \text{taxes on EBIT} + D\&A - \Delta WC - \text{net capex}$; the financing need comes from working capital and investment, not from profit alone.

Watch Out

Watch out: the numeric ABC example (WACC curve) and the Hippocrates Holding FCFO example are tables in the slides and are not recoverable from text extraction — review them on the original PDF.